Summary for 2024-08-27-ExSimplex

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December 31, 2024

1 Summary

This Linear Programming lecture focused on the simplex method, a fundamental algorithm for solving linear programming problems. The lecture progressed from introducing the basic components of a linear program to demonstrating the simplex method's iterative process, including handling cases where the origin is not feasible.

* **Linear Programming Problem:** A linear programming problem involves maximizing or minimizing a linear objective function subject to a set of linear equality or inequality constraints and non-negativity constraints on the variables.

* **Slack Variables:** Slack variables are introduced to convert inequality constraints into equality constraints, facilitating the application of the simplex method.

* **Feasible Region:** The feasible region is the set of all points that satisfy all constraints of a linear programming problem. The optimal solution always lies at a vertex of this region.

* **Simplex Method:** The simplex method is an iterative algorithm that moves from one vertex of the feasible region to another, improving the objective function value at each step until the optimal solution is found. This involves systematically interchanging basic and non-basic variables.

* **Basic and Non-basic Variables:** At each iteration of the simplex method, a subset of variables are designated as basic variables (forming a basis), while the remaining variables are non-basic variables. Setting the non-basic variables to zero defines a vertex of the feasible region.

* **Auxiliary Problem:** When the origin is not feasible, an auxiliary problem is introduced to find an initial feasible solution. This involves adding a new variable and modifying the objective function and constraints to guarantee feasibility. The original problem is feasible if and only if the optimal value of the auxiliary problem is zero.

* **Two-Phase Simplex Method:** The two-phase simplex method combines the solution of the auxiliary problem with the simplex method applied to the original problem. The auxiliary problem provides a feasible starting point for the original problem.